

Results

01 · PATH ANALYSIS

directed relationships among observed variables (path / mediation models)

Table 1. Structural paths

Path	B	SE	z	p	Std. β	95% CI	R²
x1 → x2	0.30	0.06	5.23	<.001	.30	[.19, .40]	.09
x2 → x3	0.22	0.05	4.05	<.001	.23	[.12, .34]	.24
x1 → x3	0.36	0.05	7.29	<.001	.37	[.28, .47]	.24

Table 2. Indirect effects

Path	Estimate	SE	boot 95% CI	p
x1 → x2 → x3	0.07	0.02	[0.03, 0.11]	.001

Saturation: The model is saturated (df = 0): it has zero degrees of freedom, so global fit indices are not informative and are suppressed. Estimated paths and effects below are still interpretable.

Figure. Path diagram

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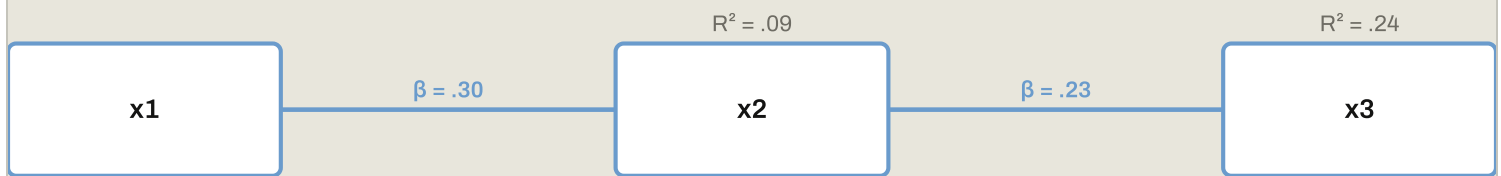
Fit

Draw path

Move

Delete





95% CI. The confidence interval for a structural path's estimate, or the bootstrap 95% CI for an indirect effect. Here, each structural path has its own 95% CI, and each indirect effect its own bootstrap 95% CI (see the tables above) - an interval excluding 0 indicates a credible effect.

R². The variance explained in each endogenous (outcome) variable. Here, R² ranges from .09 to .24 across the 3 paths.

Estimate. The size of an indirect (mediated) effect - the product of the paths through a mediator. Here, indirect-effect estimates range from 0.07 to 0.07 across the 1 mediation chain(s).

How to read this test

Path analysis estimates a system of regressions among observed variables at once, letting one variable be both an outcome and a predictor (mediation). Structural paths: each B is the unstandardized effect, Std. β the standardized effect, with z, p, and a 95% CI; R² is the variance explained in each endogenous variable. Indirect effects: the product of the paths through a mediator (e.g. $X \rightarrow M \rightarrow Y$), judged by its bootstrap 95% CI — an interval excluding 0 indicates mediation. When df = 0 the model is saturated (it reproduces the data exactly), so fit indices are not informative and are omitted; only with extra constraints (df > 0) do global fit indices apply, and even then read RMSEA cautiously at small df / small N. All thresholds are heuristics, not pass/fail gates.

APA template: A path model fit to the observed variables; the indirect effect of X on Y through M was significant, bootstrap 95% CI excluding 0.

Statistical basis: Path model estimation (lavaan::sem, observed variables only) (Rosseel, Y. (2012). "lavaan: An R Package for Structural Equation Modeling." Journal of Statistical Software, 48(2), 1-36.); RMSEA interpreted cautiously at small df / small N for over-identified models (Kenny, D. A., Kaniskan, B., McCoach, D. B. (2015). "The performance of RMSEA in models with small degrees of freedom." Sociological Methods & Research, 44(3), 486-507.); Bootstrap percentile CIs for indirect (mediated) effects (MacKinnon, D. P., Lockwood, C. M., Williams, J. (2004). "Confidence limits for the indirect effect: Distribution of the product and resampling methods." Multivariate Behavioral Research, 39(1), 99-128.). Full references in the exported CITATIONS.txt.

